

Savitribai Phule Pune University

T. Y. B.Sc. (Cyber and Digital Science) Semester V

Practical Examination CDS-355-Lab on CDS-352

Duration: 3Hrs.

Max Marks: 35



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T.Y. B.Sc. (Cyber and Digital Science)

Semester – V

C.B.C.S 2020 Pattern

CDS-355 Lab on CDS-352

(Cyber Threat Intelligence)

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Q.1) Use a threat intelligence platform (e.g., MISP or ThreatConnect) to collect and analyze indicators of compromise (IoCs) related to a recent cyber attack. [15]

Q.2) Perform a WHOIS lookup on a suspicious domain and analyze its registration details for potential threat indicators. [10]

Q.3) Viva – Questions about IoCs and the role of WHOIS in threat investigation. [5]

Q.4) Lab Book – Document the process of analyzing IoCs using a threat intelligence platform. [5]

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Q.1) Use VirusTotal to analyze a suspicious file or URL and summarize its threat intelligence data. [15]

Q.2) Perform OSINT (Open Source Intelligence) on a given IP address to identify any known threats associated with it. [10]

Q.3) Viva – Questions about VirusTotal and the role of OSINT in cyber threat intelligence. [5]

Q.4) Lab Book – Document the process of analyzing a file/URL using VirusTotal. [5]

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- Q.1)** Use Shodan to identify vulnerabilities in a specific IP address or range of IPs and report the findings. [15]
- Q.2)** Perform passive DNS analysis to track the historical domain resolution for a given malicious domain. [10]
- Q.3)** Viva – Questions about Shodan and DNS analysis in threat intelligence. [5]
- Q.4)** Lab Book – Document the steps of vulnerability identification using Shodan. [5]

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Q.1) Analyze a threat actor profile using a threat intelligence database (e.g., MITRE ATT&CK) and identify their tactics, techniques, and procedures (TTPs). [15]

Q.2) Use Maltego to visualize relationships between various entities (domains, IPs, emails) and create a threat intelligence report. [10]

Q.3) Viva – Questions on TTPs and the use of Maltego in cyber threat intelligence. [5]

Q.4) Lab Book – Record the threat actor profile analysis. [5]

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Q.1) Track and analyze phishing campaigns using Threat Intelligence feeds and report on recurring patterns or key indicators. [15]

Q.2) Investigate suspicious email headers and extract information to determine if the email is part of a known phishing attempt. [10]

Q.3) Questions about phishing campaigns and how email headers help in intelligence gathering. [5]

Q.4) Lab Book – Document the phishing campaign analysis. [5]

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Q.1) Perform domain analysis using tools like RiskIQ to identify threat actors and infrastructure associated with a malicious domain. [15]

Q.2) Use GreyNoise to filter out noise from real malicious threats based on network traffic data. [10]

Q.3) Questions about domain analysis and the role of GreyNoise in identifying true threats. [5]

Q.4) Lab Book – Document the domain analysis using RiskIQ. [5]

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- Q.1)** Use a Threat Intelligence Platform (TIP) to correlate threat data from multiple sources and generate actionable insights. [15]
- Q.2)** Perform OSINT to gather intelligence on a threat actor's online activities and summarize the findings. [10]
- Q.3)** Questions about using TIPs and OSINT techniques in cyber threat intelligence. [5]
- Q.4)** Record the threat correlation and OSINT findings. [5]

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Q.1) Use AlienVault's Open Threat Exchange (OTX) to investigate a recent cyber threat and list the IoCs associated with it. [15]

Q.2) Analyze network traffic logs using a tool like Splunk to identify suspicious patterns or potential threats. [10]

Q.3) Viva – Questions about using OTX for threat intelligence and network traffic analysis. [5]

Q.4) Lab Book – Document the analysis of threat indicators from OTX. [5]

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- Q.1)** Analyze a malware sample using online sandboxes (e.g., Any.Run or Cuckoo) to extract threat intelligence data such as network connections, file modifications, and registry changes. [15]
- Q.2)** Use a geo-IP tool to trace the origin of an attack and correlate it with known threat actors from a particular region. [10]
- Q.3)** Viva – Questions about malware analysis and the importance of geo-IP data in threat intelligence. [5]
- Q.4)** Lab Book – Record the malware analysis using a sandbox environment. [5]

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- Q.1)** Investigate and analyze threat feeds (e.g., AbuseIPDB, Spamhaus) to compile a list of malicious IPs associated with recent attacks. [15]
- Q.2)** Use the Cyber Kill Chain model to map out the stages of a real-world cyber attack and identify the attacker's methods. [10]
- Q.3)** Viva – Questions about threat feeds and the Cyber Kill Chain framework. [5]
- Q.4)** Lab Book – Document the investigation of threat feeds and use of the Cyber Kill Chain model. [5]

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Q.1) Use the MITRE ATT&CK framework to analyze an advanced persistent threat (APT) and identify its TTPs. [15]

Q.2) Analyze a suspicious URL using ThreatCrowd and determine its threat level based on associated IPs, domains, and files. [10]

Q.3) Viva – Questions about APT analysis and using ThreatCrowd in threat intelligence. [5]

Q.4) Lab Book – Document the APT analysis using MITRE ATT&CK. [5]

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Q.1) Use a threat intelligence dashboard (e.g., ThreatMiner) to track the timeline of a cyber attack and generate a report on its evolution. [15]

Q.2) Perform analysis of a DNS query log to detect any signs of DNS tunneling or exfiltration. [10]

Q.3) Viva – Questions on tracking cyber attack timelines and DNS tunneling analysis. [5]

Q.4) Lab Book – Record the steps for analyzing DNS logs for threats. [5]

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Q.1) Use Threat Intelligence feeds to detect malicious file hashes and analyze their spread across multiple campaigns. [15]

Q.2) Investigate suspicious HTTP requests using a packet capture tool (e.g., Wireshark) and determine if they are linked to any known cyber threats. [10]

Q.3) Viva – Questions about detecting malicious file hashes and analyzing HTTP requests. [5]

Q.4) Lab Book – Document the analysis of suspicious HTTP requests. [5]

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- Q.1)** Investigate known ransomware groups using a threat intelligence platform and identify common TTPs used by these groups. [15]
- Q.2)** Perform a reverse IP lookup on a malicious domain and correlate the results with other known malicious infrastructures. [10]
- Q.3) Viva** – Questions on ransomware group analysis and reverse IP lookup. [5]
- Q.4) Lab Book** – Document ransomware group intelligence and reverse IP analysis. [5]

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Q.1) Use a threat intelligence platform to gather intelligence on a data breach and analyze the data types targeted by the attackers. [15]

Q.2) Analyze SSL certificates of suspicious websites to identify signs of fraud or illegitimacy. [10]

Q.3) Viva – Questions about data breach investigations and SSL certificate analysis. [5]

Q.4) Lab Book – Document the data breach analysis using a threat intelligence platform. [5]

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Q.1) Perform a correlation analysis of multiple threat intelligence sources to identify a coordinated cyber attack campaign. [15]

Q.2) Use OSINT to track the activities of a known threat actor across various social media and underground forums. [10]

Q.3) Viva – Questions about correlation analysis in cyber threat intelligence. [5]

Q.4) Lab Book – Document the process of correlating intelligence from different sources. [5]

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Q.1) Use a Threat Intelligence Platform to identify emerging threats and analyze how they are evolving in the cyber landscape. [15]

Q.2) Perform a passive reconnaissance using domain and IP reputation tools to gather intelligence on a suspicious website. [10]

Q.3) Viva – Questions about tracking emerging threats and passive reconnaissance. [5]

Q.4) Lab Book – Record the findings on emerging threats using TIP. [5]

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- Q.1)** Analyze a phishing kit using cyber threat intelligence tools to uncover the threat actor's infrastructure and methods. [15]
- Q.2)** Perform threat intelligence gathering on a dark web marketplace and identify any illicit activity related to cyber attacks. [10]
- Q.3) Viva** – Questions on phishing kits and dark web intelligence gathering. [5]
- Q.4) Lab Book** – Document the findings of the phishing kit analysis. [5]

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Q.1) Investigate a large-scale DDoS attack using threat intelligence tools and identify the botnet behind the attack. [15]

Q.2) Perform IP reputation analysis to investigate whether certain IPs are associated with known DDoS attack campaigns. [10]

Q.3) Viva – Questions about DDoS analysis and IP reputation. [5]

Q.4) Lab Book – Document the investigation of the DDoS attack and IP reputation findings. [5]

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Q.1) Use CTI tools to perform real-time threat hunting based on IoCs obtained from recent threat reports. [15]

Q.2) Perform a reverse DNS lookup to investigate any suspicious domains linked to a malware campaign. [10]

Q.3) Viva – Questions about real-time threat hunting and DNS investigations. [5]

Q.4) Lab Book – Document the real-time threat hunting process. [5]

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Q.1) Investigate insider threats using threat intelligence tools and correlate any suspicious activity with known threat indicators. [15]

Q.2) Analyze firewall logs to detect anomalous activity indicative of insider or external threats. [10]

Q.3) Viva – Questions about insider threats and log analysis. [5]

Q.4) Lab Book – Document the findings of the insider threat investigation. [5]

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- Q.1)** Use a threat intelligence database to investigate a threat actor's infrastructure and identify linked domains and IP addresses. [15]
- Q.2)** Analyze the traffic from a malicious email attachment using a sandbox to generate actionable threat intelligence. [10]
- Q.3) Viva** – Questions about threat actor infrastructure analysis and sandbox analysis of attachments. [5]
- Q.4) Lab Book** – Document the infrastructure analysis using a threat intelligence database. [5]

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Q.1) Use a CTI platform to monitor threats targeting specific industries (e.g., finance, healthcare) and identify patterns. [15]

Q.2) Analyze threat data from multiple sources and prioritize them based on risk levels for a given industry. [10]

Q.3) Viva – Questions about industry-specific threat monitoring and prioritization techniques. [5]

Q.4) Lab Book – Record findings on industry-specific threats using a CTI platform. [5]

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Q.1) Perform threat attribution using CTI tools to determine the potential origin of a cyber attack and its actors. [15]

Q.2) Analyze domain reputation scores using various threat intelligence platforms and identify any high-risk domains. [10]

Q.3) Questions on threat attribution and domain reputation analysis. [5]

Q.4) Lab Book – Document the steps for performing threat attribution using CTI tools. [5]

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- Q.1)** Use threat intelligence feeds to analyze recent zero-day vulnerabilities and assess their potential impact. [15]
- Q.2)** Investigate suspicious user behavior using log data and correlate it with known cyber attack patterns. [10]
- Q.3) Viva** – Questions on zero-day vulnerability tracking and user behavior analysis. [5]
- Q.4) Lab Book** – Record findings from zero-day vulnerability analysis. [5]